MURSHID R

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Software Engineer • Applied AI & LLM Systems • Full-Stack Development

PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Science & AI) student at Dr. M.G.R Educational and Research Institute with strong software engineering fundamentals and hands-on experience building production-grade applications. Built SENTINEL — a full-stack LLM platform (FastAPI + React 19 + Groq) submitted to AMD Slingshot Hackathon 2026. Developed Vynta — an AI-powered Android task scheduler using Jetpack Compose, Groq API, and Google Calendar integration. Proficient in Python, JavaScript, TypeScript, REST API design, and full-stack deployment. Seeking internships in Software Engineering, Applied AI, or ML Engineering.

TECHNICAL SKILLS

Languages: Python (Intermediate) · TypeScript · JavaScript · Java · Kotlin · C++ · C · SQL · HTML/CSS
Frontend: React 19 · Next.js · Vite · Tailwind CSS · Framer Motion · TanStack Query · Jetpack Compose (Android)
Backend & APIs: FastAPI · Flask · Node.js · REST API Design · WebSockets · Celery · JWT Authentication · OAuth 2.0 · Uvicorn
AI / ML: PyTorch · TensorFlow · Keras · Scikit-learn · HuggingFace Transformers · LangChain · Groq API · Llama 3.3 70B
LLM & AI Architecture: RAG · MCP · LLM Prompting · Agentic Systems · LLM Inference Pipelines · Prompt Engineering
Databases & Storage: Supabase · PostgreSQL · SQLite · Room DB (Android) · Redis (Upstash) · Firebase
Tools & Platforms: Git/GitHub · Linux/WSL · Vercel · Render · Android Studio · Figma · Jupyter · Postman · MATLAB
ML Architectures: TCN · LSTM · CNN · Bayesian Optimization · Collaborative Filtering · Sentiment Analysis

PROJECTS

SENTINEL — Execution Intelligence Platform | AMD Slingshot Hackathon 2026 | Quantumstacks Lab *2026*

- Full-stack LLM platform that extracts delivery commitments from Slack and Gmail, scores slip risk 0–100, and auto-alerts owners before deadlines are missed
- Frontend: React 19 + TypeScript + Tailwind CSS + Framer Motion + TanStack Query — interactive dependency graph and real-time Execution Health Score dashboard
- Backend: FastAPI + Celery + Upstash Redis task queue + Supabase PostgreSQL with Row-Level Security + JWT auth
- AI pipeline: Groq LPU → Llama 3.3 70B → structured JSON extraction → 4-factor risk engine (deadline, dependencies, owner silence, workload)
- OAuth 2.0 integration: Slack, Gmail, GitHub; deployed on Vercel (frontend) + Render (backend)

Stack: Python · FastAPI · React 19 · TypeScript · Tailwind CSS · Groq · Llama 3.3 70B · Supabase · Redis · Vercel · Render

Vynta — AI Task Scheduler for Android | Personal Project *2026*

- Android app that lets users describe tasks in plain language — AI automatically schedules them based on energy level and available calendar slots
- Built with Jetpack Compose for modern declarative UI; Groq API (Llama 3.3 70B) for natural language task parsing and intelligent scheduling
- Google Calendar API integration for real-time slot detection; Room DB for local task persistence; Chronotype Architect feature aligns tasks with user productivity peaks
- Features: Neural Voice Persona, Focus Guard (silence distractions during deep work), Dynamic Gap Logic (smart task transitions), Energy Window scheduling
- Currently in testing; used daily by the developer; open beta via Google Play testing console

Stack: Kotlin · Jetpack Compose · Groq API · Google Calendar API · Room DB · Android Studio

AI-Driven Combustion Instability Prediction | Dr. M.G.R ACS Space Technology Centre *2025*

- TCN model: 92% accuracy for early warning in hybrid rocket engines; outperformed LSTM baselines
- Bayesian Optimization pipeline: 35% reduction in manual experimentation; 8-channel sensor fusion at 1000 Hz
- Presented at YASSC 2025 State Level; awarded 2nd Place and Certificate of Appreciation

Stack: Python · PyTorch · Scikit-optimize · NumPy · MATLAB

DocuMind — Mental State Prediction | Deep Learning Research *2024–2025*

- Stacked LSTM for time-series classification of mental health states from 7-day behavioral metrics; 81% Recall on At_Risk class

Stack: Python · TensorFlow/Keras · LSTM · Scikit-learn · Pandas

AIDEN AI — Intelligent Conversational Assistant | Space Club Initiative *2025*

- End-to-end conversational AI with NLP mood detection and RAG-style retrieval for adaptive academic guidance; 88% user satisfaction in beta

Stack: Python · HuggingFace Transformers · Flask API · SQLite

Ground Station Monitoring Software — Real-Time Telemetry | CanSat & Rocket Avionics *2024–2025*

- Production-grade dashboard: 8+ sensor channels at 1000 Hz, <100ms latency via WebSockets; selected for INSPACe national competition

Stack: Python · FastAPI · WebSockets · Plotly · SQLite

Enterprise Data Automation & Analytics Pipeline | Personal Project *2025–2026*

- Automated Python pipeline cleaning messy CSV exports with Pandas + Regex; real-time Streamlit BI dashboard with one-click export

Stack: Python · Pandas · NumPy · Plotly Express · Streamlit

Smart Attendance System — Desktop Application | Personal Project *2025–2026*

- Dual-mode desktop app: Admin Dashboard + Staff Terminal (PIN clock-in); packaged as standalone Windows .exe; SQLite local storage

Stack: Python · CustomTkinter · SQLite · Pandas

PROFESSIONAL EXPERIENCE

AI Research Engineer | Dr. M.G.R ACS Space Technology Centre, Chennai *2024 – Present*

- Developed TCN model achieving 92% accuracy in real-time combustion instability prediction; engineered Bayesian Optimization pipeline reducing manual iterations by 35%
- Architected digital twin workflow integrating 8-channel sensor data fusion at 1000 Hz; reduced chamber pressure prediction error by 18%
- Engineered payload integration workflows for CubeSat and rocketry missions; 100% compliance with aerospace safety protocols

Mathematics & Chemistry Instructor | B.L Learning Institute, Chennai *2023 – 2024*

- Taught 50+ students; achieved 15–20% average performance improvement; designed 40+ interactive problem sets using Figma

EDUCATION

B.Tech — Computer Science & Engineering (Data Science & AI Specialization) | Dr. M.G.R Educational and Research Institute, Chennai *2023 – Present*

- GPA: 8.60/10.00 (Semester 4) | Upward trajectory: 7.38 → 7.71 → 8.14 → 8.60
- Relevant Coursework: Data Structures & Algorithms · Machine Learning · Deep Learning · Operating Systems · Computer Networks · Signal Processing · Databases

AWARDS & ACHIEVEMENTS

- AMD Slingshot Hackathon 2026 — SENTINEL Submission — Quantumstacks Lab *2026*
- 2nd Place — Paper Presentation (YASSC 2025) — Mechanical Engineering Department *Sep 2025*
- Certificate of Appreciation — Research Paper — State YASSC 2025 *Aug 2025*

- Runner-Up — Promptarhon AI Competition — AI: Beyond the Algorithm *May 2025*
- Selected Team — National Rocketry Competition — INSPACe Model Rocketry India *Oct 2025*
- TiHAN Program Participant — IIT Hyderabad — Autonomous Navigation & Robotics *2024*

LANGUAGES

English (Fluent) · Tamil (Native) · Hindi (Conversational) · German (Basic)